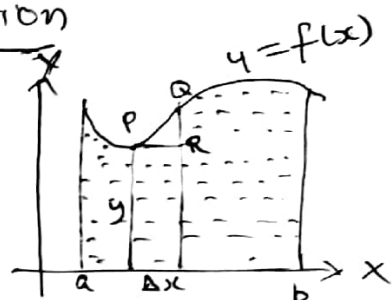


- (6) Find the length of the curve $x=t^2$, $y=t^3$ between $t=0$ and $t=1$
- (7) Find the length of one arch of the cycloid $x=a(\theta + \sin\theta)$, $y=a(1-\cos\theta)$
- (8) Find the perimeter of the cardioid
• $r = 2(1-\cos\theta)$

Volume of solid of revolution

Consider area bounded by the curve $y=f(x)$, the x -axis and the ordinates at $x=a$ and $x=b$; when this area is revolved about x -axis a solid will be generated. We wish to find volume of this solid formed.



Consider a small strip at $p(x,y)$ of width Δx . When this strip is revolved about x -axis a circular disc of radius y and thickness Δx will be formed.

$$\text{Volume of this circular disc} = \pi y^2 \Delta x = \pi y^2 dx$$

$\therefore$ As x varies from a to b volume of the solid generated is given by

$$V = \int_a^b \pi y^2 dx$$

Note The volume of the solid formed by revolution of the area bounded by $x=g(y)$, the y -axis and the abscissa at $y=c$ and $y=d$ is given by

$$V = \int_c^d x^2 dy$$

(2) Parametric form

Volume of the solid obtained by revolving the part of the curve $x = f(t)$, $y = g(t)$ bounded by x -axis and the lines $t = t_1$ and $t = t_2$ is given by ^{revolved about x -axis}

$$V = \int_{t_1}^{t_2} \pi y^2 \frac{dx}{dt} dt$$

(3) Polar form

Volume of the solid generated by revolving the area bounded by the curve $r = f(\theta)$ and the radii vectors $\theta = \alpha$ and $\theta = \beta$

(i) about the initial line ($\theta = 0$)

$$V = \int_{\alpha}^{\beta} \frac{2}{3} \pi r^3 \sin \theta d\theta$$

(ii) about the line $\theta = \frac{\pi}{2}$

$$V = \int_{\alpha}^{\beta} \frac{2}{3} \pi r^3 \cos \theta d\theta$$

(iii) about the line $\theta = \gamma$

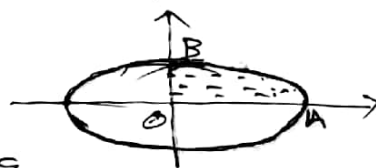
$$V = \int_{\alpha}^{\beta} \frac{2}{3} \pi r^3 \sin(\theta - \gamma) d\theta$$

Examples

(i) Find the volume of the solid formed by revolution about major axis of the ellipse with axes $2a$ and $2b$

The eq of the ellipse may be taken as $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

The major axis lie along x -axis, assuming that $a > b$



Since this ellipse is symmetrical about both the axes, required volume is double the volume of the solid obtained by revolving the area OAB (shaded in the figure) in the first quadrant

For the arc BA x -values from 0 to a

$$\therefore \text{Required volume} = 2 \times \int_0^a \pi y^2 dx$$

$$= 2\pi \int_0^a y^2 dx$$

$$= 2\pi \int_0^a \frac{b^2}{a^2} (a^2 - x^2) dx$$

$$= \frac{2\pi b^2}{a^2} \int_0^a (a^2 - x^2) dx$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\text{or } \frac{y^2}{b^2} = 1 - \frac{x^2}{a^2}$$

$$\therefore y^2 = \frac{b^2}{a^2} (a^2 - x^2)$$

$$= \frac{2\pi b^2}{a^2} \left[a^2 x - \frac{x^3}{3} \right]_0^a = \frac{2\pi b^2}{a^2} \left(a^3 - \frac{a^3}{3} \right)$$

$$= \frac{2\pi b^2 a^3}{a^2} \left(\frac{2}{3} \right) = \frac{4}{3} \pi a b^2$$

Note Putting $b=a$ in the above result, we obtain volume of a sphere of radius a as $\frac{4}{3} \pi a^3$

(2) Obtain the volume of the solid generated by revolving the part of the curve $x^2(y-x^2) = 3$ between $x=1$ and $x=2$ about x axis

$$\text{Given eq. is } x^2(y-x^2) = 3 \Rightarrow y-x^2 = \frac{3}{x^2}$$

$$\Rightarrow y = x^2 + \frac{3}{x^2} \text{ limits of integration are 1 and 2}$$

$$\text{Required volume} = \int_1^2 \pi y^2 dx$$

$$= \pi \int_1^2 \left(x^2 + \frac{3}{x^2} \right)^2 dx$$

$$= \pi \int_1^2 \left(x^4 + 6 + \frac{9}{x^4} \right) dx$$

$$= \pi \left[\frac{x^5}{5} + 6x + 9 \frac{x^{-3}}{-3} \right]_1^2$$

$$= \pi \left[\frac{x^5}{5} + 6x - \frac{3}{x^3} \right]_1^2$$

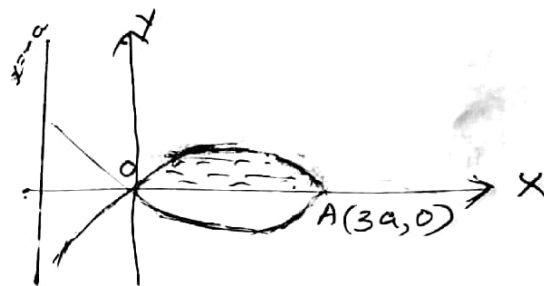
$$= \pi \left[\left(\frac{32}{5} + 12 - \frac{3}{8} \right) - \left(\frac{1}{5} + 6 - 3 \right) \right]$$

$$= \pi \left[\frac{31}{5} + 9 - \frac{3}{8} \right] = \pi \left(\frac{248 + 360 - 15}{40} \right)$$

$$= \frac{593}{40} \pi$$

Example 3 Find the volume of the solid obtained by revolving the loop of the curve $y^2(a+x) = x^2(3a-x)$ about x-axis

Since the eq of the curve contains ^{only} even power terms (i.e. y^2) it is symmetric about



x-axis. Its intersection with x-axis is got by putting $y=0 \Rightarrow x^2(3a-x)=0 \Rightarrow x=0, x=3a$. Hence the curve has a loop lying between $x=0$ and $x=3a$

Since the loop is symmetric about x-axis, volume generated by revolving the loop is same as the volume generated revolving part of the loop above x-axis

∴ Required volume is

$$\begin{aligned}
 V &= \int_0^{3a} \pi y^2 dx \\
 &= \pi \int_0^{3a} \frac{x^2(3a-x)}{x+a} dx \\
 &= \pi \int_0^{3a} \frac{3ax^2 - x^3}{x+a} dx
 \end{aligned}$$

Dividing $-x^3 + 3ax^2$ by $x+a$
 quotient = $-x^2 + 4ax - 4a^2$
 remainder = $4a^3$

$$\therefore \frac{3ax^2 - x^3}{x+a} = -x^2 + 4ax - 4a^2 + \frac{4a^3}{x+a}$$

$$\begin{array}{r}
 -x^2 + 4ax - 4a^2 \\
 x+a \overline{) -x^3 + 3ax^2} \\
 \underline{-x^3 - ax^2} \\
 4ax^2 \\
 \underline{4ax^2 + 4a^2x} \\
 -4a^2x - 4a^3 \\
 \underline{-4a^2x - 4a^3} \\
 4a^3
 \end{array}$$

$$\begin{aligned}
 \therefore V &= \pi \int_0^{3a} \left(-x^2 + 4ax - 4a^2 + \frac{4a^3}{x+a} \right) dx \\
 &= \pi \left[-\frac{x^3}{3} + 2ax^2 - 4a^2x + 4a^3 \log(x+a) \right]_0^{3a} \\
 &= \pi \left[\left(-\frac{27a^3}{3} + 2a(9a^2) - 4a^2(3a) + 4a^3 \log 4a \right) \right. \\
 &\quad \left. - 4a^3 \log a \right] \\
 &= \pi \left[-9a^3 + 18a^3 - 12a^3 + 4a^3 (\log 4a - \log a) \right] \\
 &= \pi \left[-3a^3 + 4a^3 \log 4 \right] \\
 &= \pi a^3 \left[-3 + 4 \log 2^2 \right] = \underline{\underline{\pi a^2 [8 \log 2 - 3]}}
 \end{aligned}$$

(4) Find the volume of the solid obtained by revolving one arch of the cycloid

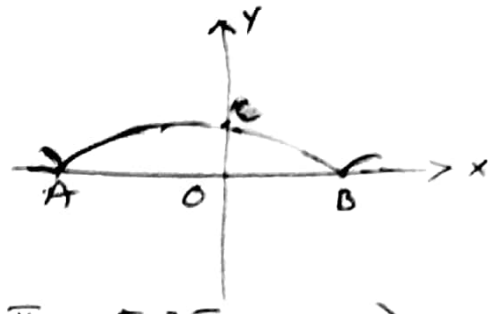
$x = a(\theta + \sin \theta)$, $y = a(1 + \cos \theta)$ about x -axis.

The curve meets the
x-axis when $y=0$

$$\Rightarrow a(1+\cos\theta) = 0$$

$$\Rightarrow \cos\theta = -1$$

$$\text{i.e. } \theta = (2n-1)\pi \quad (\text{i.e. } \theta = \pm\pi, \pm 3\pi, \dots)$$



putting $n=0$ and 1 we get two consecutive points of intersection at A and B where $\theta = -\pi$ and $\theta = \pi$ respectively. Also this curve is symmetric about y axis

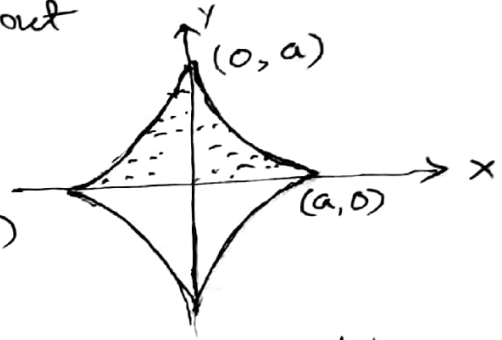
$$\begin{aligned} \text{Volume generated} &= \int_{-\pi}^{\pi} \pi y^2 dx \\ &= 2 \int_0^{\pi} \pi y^2 \frac{dx}{d\theta} d\theta \\ &= 2\pi \int_0^{\pi} a^2 (1+\cos\theta)^2 \cdot a(1+\cos\theta) d\theta \\ &= 2\pi a^3 \int_0^{\pi} (2\cos^2 \frac{\theta}{2})^3 d\theta \\ &= 2\pi a^3 \int_0^{\pi} 8 \cos^6 \frac{\theta}{2} d\theta \\ &= 16\pi a^3 \int_0^{\pi} \cos^6 \frac{\theta}{2} d\theta \\ &= 16\pi a^3 \int_0^{\pi/2} \cos^6 \phi (2d\phi) \\ &= 32\pi a^3 \int_0^{\pi/2} \cos^6 \phi d\phi \\ &= 32\pi a^3 \cdot \frac{5}{6} \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} \\ &= \underline{5\pi^2 a^3} \end{aligned}$$

put $\phi = \frac{\theta}{2}$
 $d\phi = \frac{d\theta}{2}$
 when $\theta = \pi$, $\phi = \frac{\pi}{2}$
 when $\theta = 0$, $\phi = 0$

- (5) Find the volume of solid of revolution obtained by revolving the curve $x^{2/3} + y^{2/3} = a^{2/3}$ about x-axis

In the parametric form $x = a \cos^3 \theta$, $y = a \sin^3 \theta$
 The curve is symmetric about

both axes. It meets the
 x -axis at $(a, 0)$ and $(-a, 0)$
 and y -axis at $(0, a)$ and $(0, -a)$



Volume generated

= twice the volume generated by
 revolving the portion in the
 I quadrant about x -axis

Volume in the first quadrant is bounded by

$$\theta = \frac{\pi}{2} \text{ and } \theta = 0$$

$$\therefore V = 2 \int_{\pi/2}^0 \pi y^2 \frac{dx}{dt} dt$$

$$= 2\pi \int_{\pi/2}^0 a^2 \sin^6 \theta (-3a \cos^2 \theta \sin \theta) d\theta$$

$$= -6\pi a^3 \int_{\pi/2}^0 \sin^7 \theta (\cos^2 \theta) d\theta$$

$$= 6\pi a^3 \int_0^{\pi/2} \sin^7 \theta (1 - \sin^2 \theta) d\theta$$

$$= 6\pi a^3 \left[\int_0^{\pi/2} \sin^7 \theta d\theta - \int_0^{\pi/2} \sin^9 \theta d\theta \right]$$

$$= 6\pi a^3 \left(\frac{6}{7} \cdot \frac{4}{5} \cdot \frac{2}{3} \times 1 - \frac{8}{9} \times \frac{6}{7} \times \frac{4}{5} \times \frac{2}{3} \times 1 \right)$$

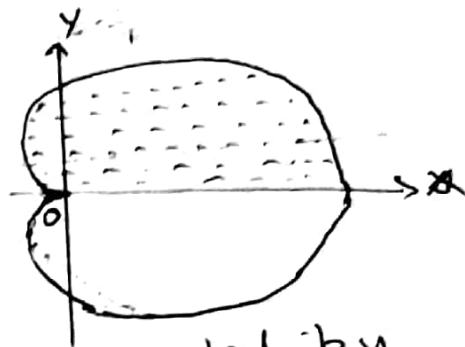
$$= 6\pi a^3 \left(\frac{6}{7} \times \frac{4}{5} \times \frac{2}{3} \left(1 - \frac{8}{9} \right) \right)$$

$$= 6\pi a^3 \cdot \frac{6}{7} \times \frac{4}{5} \times \frac{2}{3} \times \frac{1}{9}$$

$$= \frac{32}{105} \pi a^3$$

- (6) Find the volume of the solid obtained by
 revolving the cardioid $r = a(1 + \cos \theta)$
 about the initial line.

The curve $r = a(1 + \cos \theta)$ is symmetric about the initial line. The volume generated by revolving the cardioid is same as the volume of the solid generated by revolving the upper half of the cardioid.



$$\begin{aligned}
 \text{Required volume} &= \int_0^\pi \frac{2}{3} \pi r^3 \sin \theta \, d\theta \\
 &= \frac{2}{3} \pi \int_0^\pi a^3 (1 + \cos \theta)^3 \sin \theta \, d\theta \\
 &= \frac{2}{3} \pi a^3 \left[\frac{(1 + \cos \theta)^4}{-4} \right]_0^\pi \\
 &= \frac{2}{3} \pi a^3 \left(\frac{0 - 2^4}{-4} \right) \\
 &= \frac{32}{3} \pi a^3
 \end{aligned}$$

Problems

- (1) Find the volume of the solid generated by revolving the area of the hyperbola $y^2 = 4ax$ bounded by the ordinate at $x = h$ about x-axis.
- (2) Find the volume of the solid generated by revolving the part of the parabola $y^2 = 4ax$ cut off by the Latusrectum is rotated about its axis.
- (3) S.T. volume of the solid generated by the revolution of the loop of the curve $x = t^2$, $y = t - \frac{t^3}{3}$ about x-axis is $\frac{3\pi}{4}$.
- (4) Find the volume of the solid generated by revolving one loop of the curve $r^2 = a^2 \cos 2\theta$ about the line $\theta = \frac{\pi}{2}$.